

Surface-Code Syndrome Utilization for Fault Identification

Can repeated stabilizer syndromes reveal *where* a device is faulty—not just *that* it is?

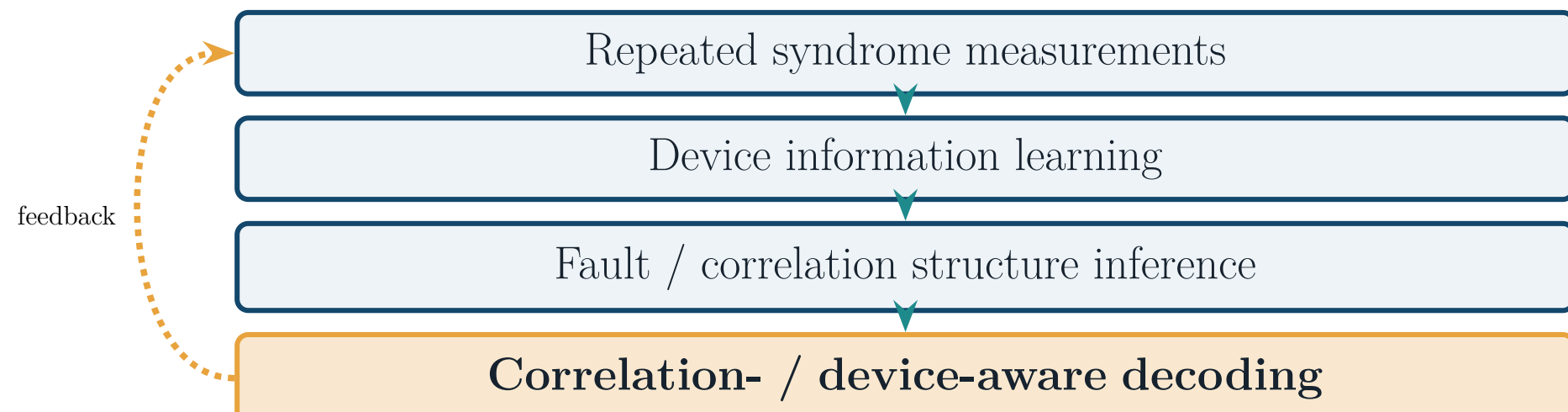
Jaehun Han¹ [co-authors / advisor placeholder]¹ [Institution / Group placeholder]
pjhan.research@gmail.com • Niels Bohr Quantum Summer School 2026

1. Motivation

Surface codes already measure syndromes every round—thousands of stabilizer bits are produced purely to detect and decode errors, then discarded.

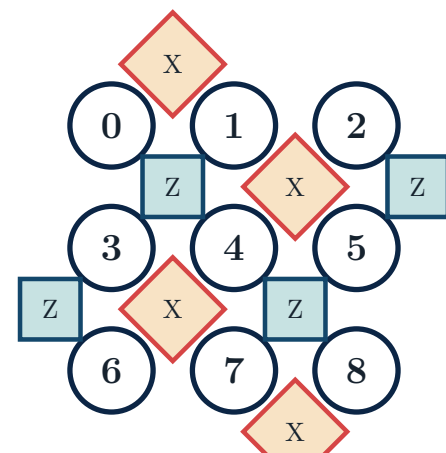
“This syndrome stream is a free probe of the device. Can we read its own fault structure out of data we already collect for QEC?”

Long-term vision: a **closed loop** that recycles QEC data into device knowledge, then feeds it back into better decoding.



2. Case Study & Problem

Setting. Distance-3 rotated surface code (17 qubits: 9 data + 8 ancilla); each stabilizer-extraction round issues **24 directed CNOTs**. One CNOT is the *dominant* fault source (higher fault probability than the rest).



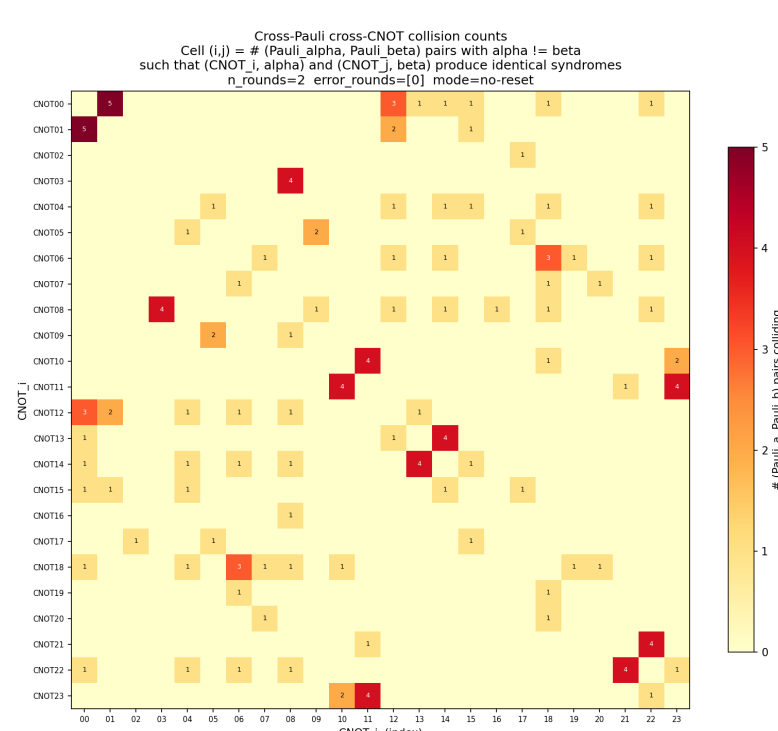
Rotated $d=3$ layout: 9 data qubits (circles), 8 Z-/X-stabilizer ancillas (squares / diamonds).

The question

From a temporal syndrome / detection-event sequence $S = (s_1, \dots, s_T)$ *alone*, identify the **dominant faulty CNOT** among the 24. **Nuisance factors** (unobserved): which rounds the fault fires, and its Pauli type ($XX, \dots, ZZ$).

3. Single-Shot Ambiguity

Is one measurement enough? **No**. When the Pauli type is unknown, distinct (CNOT, Pauli) faults can produce *identical* syndromes. Enumerating all 216 single-round cases:



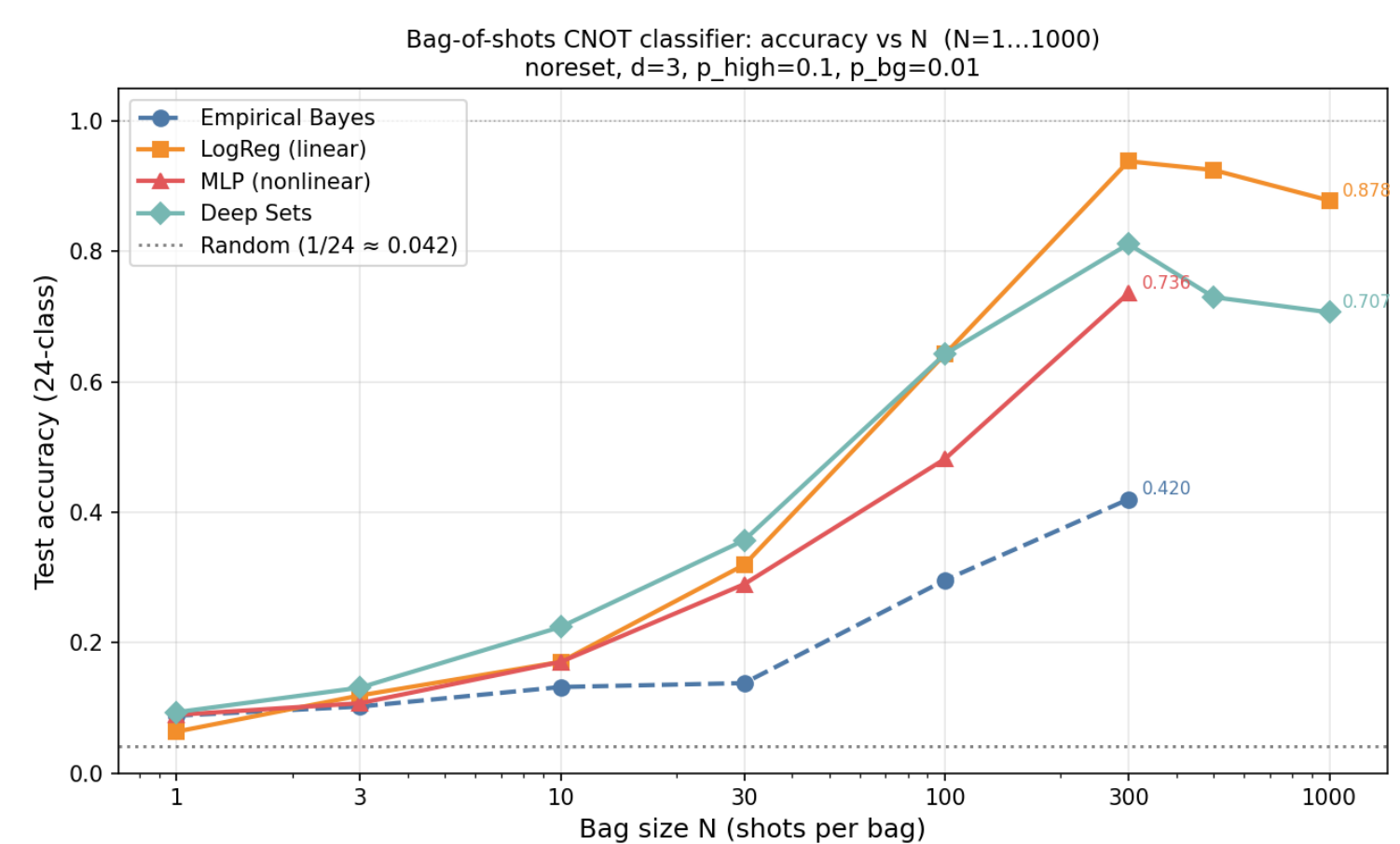
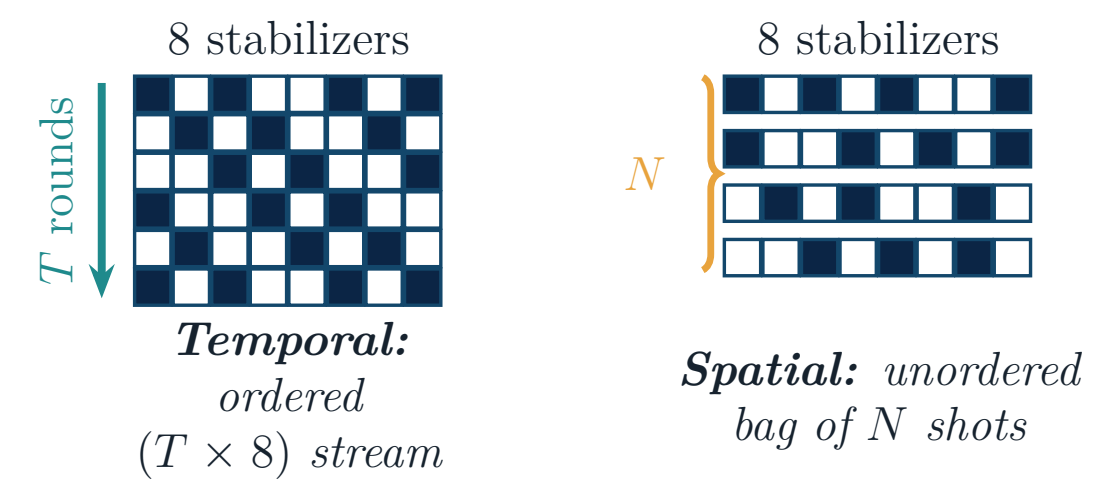
Collision map: cell (i, j) counts Pauli pairs for which faults at $CNOT_i$ and $CNOT_j$ are indistinguishable.

Hard limit

216 cases collapse to 123 signatures with **72** cross-CNOT collisions; a single round with known Pauli identifies only 8/24. **Single-shot 24-class identification is information-theoretically impossible**—invariant under round count and reset mode.

4. More Data Breaks the Ambiguity

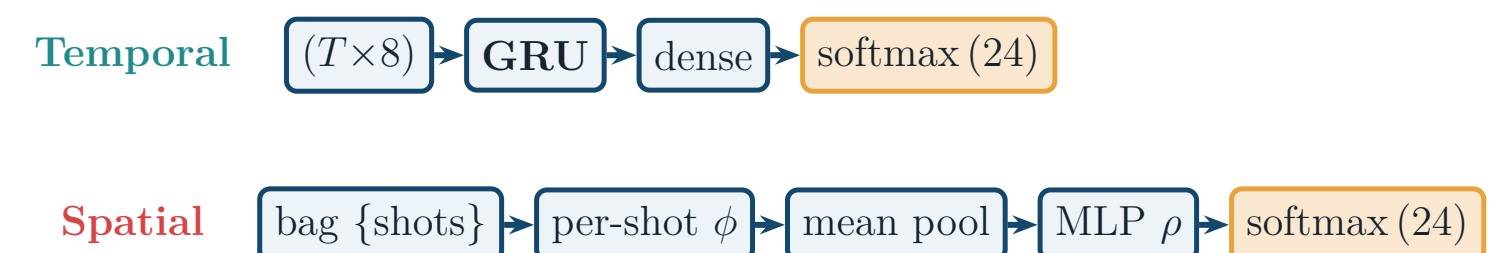
The way out is **aggregation**. The same fault is probed either *temporally* (a long syndrome stream) or *spatially* (many repeated single-round shots):



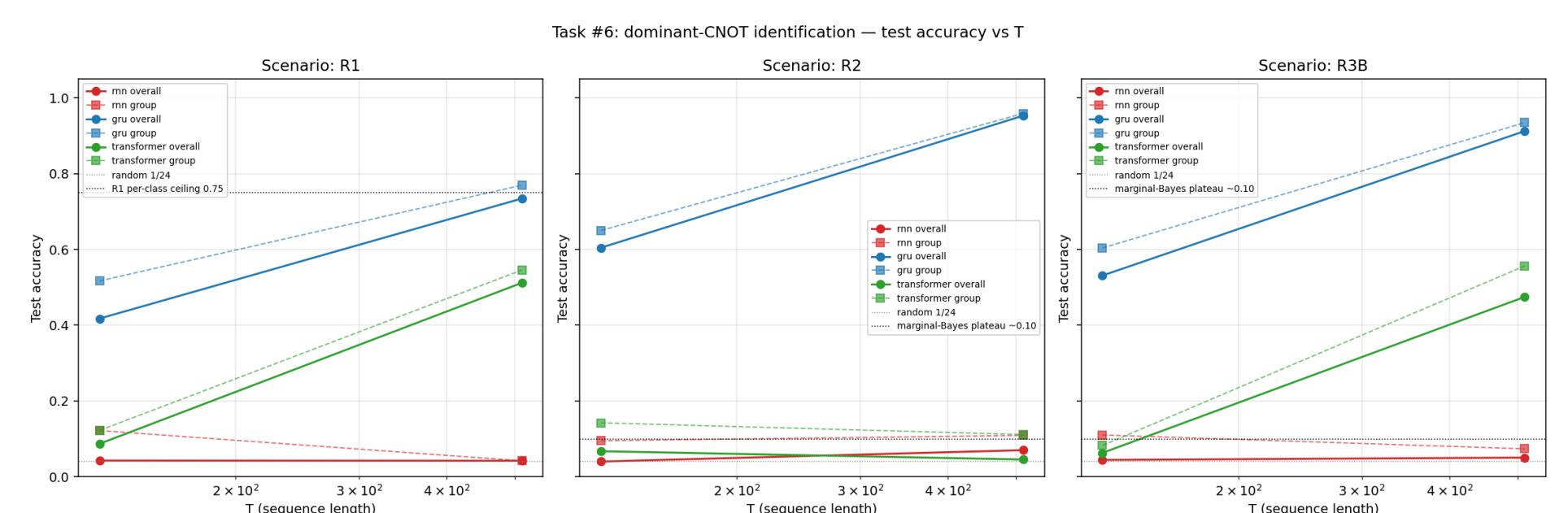
Bag-of-shots CNOT accuracy vs bag size N . At $N=1$ (single shot) accuracy is near random ($1/24$); by $N=300$ it reaches ≈ 0.9 —the ambiguity is broken by data volume, not by a better single shot.

5. Learned Decoders: Models & Performance

Each regime has a matched model class. Best performers so far (**preliminary**, naive $d=3$):



Deep Sets (spatial) is permutation-invariant over the bag; a linear LogReg on pooled features is a strong baseline.



Temporal accuracy vs sequence length T (three fault-model scenarios). The GRU (blue) climbs with T to $\approx 95\%$ (R2), breaking the rule-based single-round ceiling (0.75); RNN / transformer lag.

Best accuracy so far (preliminary, $d=3$)

$\approx 95\%$ Temporal GRU **$\approx 94\%$** Spatial LogReg ($N=300$)
Reported as in-progress, not final performance claims.

6. Status & Outlook

Done. Exact structural map of what syndromes can / cannot reveal about fault location at $d=3$.

Ongoing. Stim-based *realistic* circuit (parallel schedule, general d); *probabilistic* dominant-fault model.

Next. Close the loop—feed inferred fault / correlation structure into **device-aware decoding**.